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Collapsible element for façade systems

Abstract

A façade system includes a mullion having exterior and interior portions and defining a glazing pocket between the exterior and interior portions, a thermal break arranged within the glazing pocket and extending between the exterior and interior portions, the thermal break dividing the glazing pocket into a shallow pocket and a deep pocket larger than the shallow pocket, and a collapsible element arranged within the deep pocket and extending between the thermal break and a lateral side of a panel introduced into the deep pocket. The collapsible element is movable between a collapsed state and an expanded state. The collapsible element divides the deep pocket into two or more thermal chambers when in the expanded state to reduce heat transfer by convection through the glazing pocket.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5309689	12/1993	Croissant	52/235	E04B 2/967
2021/0355679	12/2020	Newbrough	N/A	E04B 2/92

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
29608742	12/1995	DE	E06B 1/36
19707624	12/1997	DE	E04B 2/967
20003060	12/1999	DE	E06B 3/2632
10015838	12/2000	DE	E04B 2/967
102005032176	12/2006	DE	E06B 3/26303
202014100868	12/2014	DE	E06B 3/2632
202019106571	12/2020	DE	E06B 3/22
1174554	12/2001	EP	E04B 2/967
2048315	12/2008	EP	E06B 3/26303
2631407	12/2012	EP	E06B 1/62
2941003	12/2009	FR	E06B 3/26347

OTHER PUBLICATIONS

Machine translation of EP2048315 (Year: 2024). cited by examiner

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Background/Summary

BACKGROUND

(1) Façade systems are commonly used in commercial buildings and generally comprise the structural elements that provide lateral and vertical resistance to wind and other actions, and further

include the building envelope elements that provide weather resistance and thermal, acoustic, and fire resisting properties. Storefronts, window walls, and curtain walls are often used in the exterior of high-rise buildings. The overall energy efficiency of a building, including energy transfer characteristics of its façade system, is an important factor in architectural design, and there is a continued demand for building features and methods of construction that improve energy efficiency.

(2) Some façade systems utilize frames made of metal, such as aluminum or aluminum alloy, and metal frames are particularly good thermal conductors. Thus, improved and/or alternative structures and methods for controlling the heat transfer characteristics of façade systems and for achieving aesthetic design objectives remain desirable.

SUMMARY OF THE DISCLOSURE

(3) Embodiments disclosed herein include a façade system that includes a mullion having exterior and interior portions and defining a glazing pocket between the exterior and interior portions, a thermal break arranged within the glazing pocket and extending between the exterior and interior portions, the thermal break dividing the glazing pocket into a shallow pocket and a deep pocket larger than the shallow pocket, and a collapsible element arranged within the deep pocket and extending between the thermal break and a lateral side of a panel introduced into the deep pocket. The collapsible element is movable between a collapsed state and an expanded state, and wherein the collapsible element divides the deep pocket into two or more thermal chambers when in the expanded state to reduce heat transfer by convection through the glazing pocket. In a further embodiment of the façade system, the collapsible element is naturally biased to the expanded state. In another further embodiment of the façade system, the collapsible element is naturally biased to the collapsed state. In another further embodiment of the façade system, the collapsible element includes two side walls that fold inward upon moving to the collapsed state. In another further embodiment of the façade system, the collapsible element includes two side walls that fold outward upon moving to the collapsed state. In another further embodiment of the façade system, the collapsible element includes two side walls and an inner wall interposing the two side walls, and wherein the two side walls and the inner wall divide the deep pocket into the four thermal chambers. In another further embodiment of the façade system, the side walls fold outward and the inner wall folds toward one of the side walls upon moving to the collapsed state. In another further embodiment of the façade system, the collapsible element includes two side walls and a cross-member extending between the side walls, and wherein the size walls are folded over one another when in the collapsed state. In another further embodiment of the façade system, wherein at least one of the side walls extends between the thermal break and the lateral side of the panel upon transitioning to the expanded state. In another further embodiment of the façade system, the collapsible element comprises a first portion and a second portion separate from the first portion, each portion providing a side wall securable to the mullion and interconnected with a foldable inner wall, wherein the foldable inner wall is engageable with the lateral side upon transitioning to the expanded state. In another further embodiment of the façade system, the collapsible element includes first and second foldable inner walls that divide the deep pocket into three thermal chambers upon transitioning to the expanded state. In another further embodiment of the façade system, the collapsible element further includes opposing first and second side walls, and a cross-member extending between and interconnecting the opposing first and second side walls, wherein the foldable inner walls extend from corresponding transition points where the opposing first and second side walls meet the cross-member. In another further embodiment of the façade system, the collapsible element is secured to mullion or the panel with an attachment means selected from the group consisting of an adhesive, a coupling device, an interference fit, a snap-fit engagement, and any combination thereof. In another further embodiment of the façade system, the panel comprises a first panel and the system further comprises a second panel laterally offset from the first panel, wherein the glazing pocket is defined between the exterior and interior portions of the mullion and

between lateral ends of the first and second panels, first and second exterior gaskets providing corresponding sealed interfaces between the first and second panels and the exterior portion of the mullion, and first and second interior gaskets providing corresponding sealed interfaces between the first and second panels and the interior portion of the mullion, wherein the first and second exterior and interior gaskets substantially seal the glazing pocket.
(4) Embodiments disclosed herein may further include a method of reducing heat transfer through the façade system of the previous paragraph, the method may include the steps of dividing the deep pocket of the glazing pocket into the two or more thermal chambers with the collapsible element when the collapsible element is transitioned to the expanded state, and reducing heat transfer by convection through the glazing pocket with the collapsible element in the expanded state.
(5) Embodiments disclosed herein may further include a method of assembling a façade system, the method may include coupling a first panel to a mullion, the mullion including an exterior portion and an interior portion, a glazing pocket defined between the exterior and interior portions, and a thermal break arranged within the glazing pocket and extending between the exterior and interior portions and thereby dividing the glazing pocket into a shallow pocket and a deep pocket larger than the shallow pocket, wherein the first panel is received within the shallow pocket. The method may further include advancing a second panel into the deep pocket and toward the thermal break, wherein a collapsible element is arranged in the deep pocket and movable between a collapsed state and an expanded state, and dividing the deep pocket into two or more thermal chambers with the collapsible element in the expanded state. The collapsible element is naturally biased to the expanded state and advancing the second panel into the deep pocket comprises collapsing the collapsible element to the collapsed state as the second panel advances into the deep pocket. The method may further include advancing the second panel into the second pocket at an angle offset from perpendicular to the thermal break. The method may further include drawing the second panel partially out of the deep pocket and thereby allowing the collapsible element to transition from the collapsed state to the expanded state. The collapsible element is secured to at least one of the thermal break and the lateral side of the panel with an attachment means selected from the group consisting of an adhesive, a coupling device, an interference fit, a snap-fit engagement, and any combination thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The following figures are included to illustrate certain aspects of the present disclosure, and should not be viewed as exclusive embodiments. The subject matter disclosed is capable of considerable modifications, alterations, combinations, and equivalents in form and function, without departing from the scope of this disclosure.
(2) FIG. **1** is schematic top view of a prior art façade system.
(3) FIGS. **2**A and **2**B are schematic top views of an example façade system that incorporates the principles of the present disclosure.
(4) FIGS. **3**A and **3**B are schematic top views of another example façade system, in accordance with one or more additional embodiments of the present disclosure.
(5) FIGS. **4**A and **4**B are schematic top views of another example façade system, in accordance with one or more additional embodiments of the present disclosure.
(6) FIGS. **5**A and **5**B are schematic top views of another example façade system, in accordance with one or more additional embodiments of the present disclosure.
(7) FIGS. **6**A and **6**B are schematic top views of another example façade system, in accordance with one or more additional embodiments of the present disclosure.
(8) FIGS. **7**A and **7**B are schematic top views of another example façade system, in accordance

with one or more additional embodiments of the present disclosure.

(9) FIGS. **8-11** depict example attachment means for securing collapsible element within corresponding systems.

(10) FIG. **12** is a cross-sectional view of a curtain wall system that may incorporate the principles of the present disclosure.

(11) FIGS. **13**A and **13**B are side-by-side depictions of thermal simulations of the system of FIGS. **2**A-**2**B.

DETAILED DESCRIPTION

(12) The present disclosure is related to building products and, more particularly, to collapsible elements for reducing heat transfer by convection in façade systems.

(13) Embodiments described herein disclose various designs and configurations of collapsible elements that may be arranged within glazing pockets of façade systems to help reduce convective heat transfer. The collapsible elements described herein divide the volume of air within the glazing pockets into multiple thermal chambers. This may prove advantageous in providing an inexpensive method of improving the thermal performance of façade systems. Moreover, the embodiments discussed herein may be adaptable to existing façade systems and otherwise consist in a universal method that can fit multiple façade systems.

(14) FIG. **1** is schematic top view of a prior art façade system **100**. The façade system **100** (hereafter "the system **100**") shown in FIG. **1** is an example storefront and could be applicable to large and small commercial buildings or residential buildings. The principles of the present disclosure, however, are also applicable to other types of façade systems, such as curtain wall systems, without departing from the scope of the disclosure.

(15) As illustrated, the system **100** includes a vertical mullion **102** having a first or "exterior" portion **104***a* and a second or "interior" portion **104***b*. The exterior portion **104***a* is generally exposed to the exterior of a building, while the interior portion **104***b* is generally exposed to the interior of the building. The vertical mullion **102** may comprise a rigid extrusion made of aluminum, an aluminum alloy, or other material, including, but not limited to, other metals and alloys.

(16) The vertical mullion **102** is designed to laterally support and/or secure one or more window panels, shown in FIG. **1** as a first panel **106***a* and a second panel **106***b* laterally offset from each other. The panels **106***a,b* may comprise glazing panels, but may alternatively comprise one or more panes of window glass, one or more panes of polycarbonate, or one or more panels of material that are clear, translucent, tinted, or opaque.

(17) The panels **106***a,b* are secured to the mullion **102**, at least in part, using one or more seals or gaskets, shown as exterior gaskets **108***a* and interior gaskets **108***b*. The exterior gaskets **108***a* provide a sealed interface between the panels **106***a,b* and the adjacent exterior portion **104***a* of the mullion **102**, and the interior gaskets **108***b* provide a sealed interface between the panels **106***a,b* and the adjacent interior portion **104***b* of the mullion **102**.

(18) The mullion **102** extends from the exterior to the interior and defines a glazing pocket **110** configured and sized to receive and secure the panels **106***a,b*. To improve thermal performance of the system **100**, the mullion **102** includes and otherwise provides a thermal break **112** that extends through the glazing pocket **110** and interconnects the exterior and interior portions **104***a,b*. The thermal break **112** may be made of one or more materials having a thermal conductivity that is less than a thermal conductivity of the vertical mullion **102**.

(19) The thermal break **112** may comprise any type of suitable thermal break capable of preventing conductive thermal energy loss between the exterior and interior portions **104***a,b*. In the illustrated example, the thermal break **112** comprises two interconnected pour and debridge (PND) thermal breaks consisting of a urethane material or the like. Moreover, the portions of the thermal break **112** are connected with a bridge **114**, which may be made of aluminum, for example.

(20) The thermal break **112** effectively divides the glazing pocket **110** into a first or "shallow"

pocket **116***a* and a second or “deep” pocket **116***b*. As illustrated, the mullion **102** is configured such that the shallow pocket **116***a* exhibits a smaller size or volume as compared to the deep pocket **116***b*. Inclusion of the shallow and deep pockets **116***a,b* is designed to help in the assembly or installation process of the system **100**.

(21) More specifically, the system **100** is assembled by first receiving the first panel **106***a* into the shallow pocket **116***a* and thereby securing the first panel **106***a* to the mullion **102**. The second panel **106***b* can then be advanced into the deep pocket **116***b* and situated perpendicular to the mullion **102**. The depth of the deep pocket **116***b* allows the second panel **106***b* to be initially advanced into the deep pocket **116***b* toward the thermal break **112** at an angle offset from perpendicular to the mullion **102**, which may be required due to tight manufacturing and construction tolerances and constraints. Once advanced into the deep pocket **116***b*, the orientation of the second panel **106***b* can then be adjusted to be perpendicular to the mullion **102**, following which the second panel **106***b* may then be drawn or pulled away from the thermal break **112** a small distance while still remaining within the deep pocket **116***b*. In some installations, drawing the second panel **106***b* away from the thermal break **112** within the deep pocket **116***b* can simultaneously allow the installer to advance the opposing lateral side (not shown) of the second panel **106***b* into an adjacent shallow pocket (not shown) of an adjacent vertical mullion (not shown).

(22) While the deep pocket **116***b* can serve an essential role during installation and assembly of the system **100**, a large volume of air remains in the deep pocket **116***b* following installation. This can contribute to undesireable heat transfer by convection through the glazing pocket **110**, and heat transfer by convection through the deep pocket **116***b* will negatively affect the thermal performance of the system **100**.

(23) According to embodiments of the present disclosure, the thermal performance of the system **100** may be improved by including or otherwise installing a collapsible element within the deep pocket **116***b* and generally arranged between the thermal break **112** and an adjacent lateral side **118** of the second panel **106***b*. The collapsible element may be designed to divide the deep pocket **116***b* into two or more thermal chambers, which correspondingly divides the volume of air within the deep pocket **116***b* and thereby operates to reduce heat transfer by convection through the glazing pocket **110**.

(24) FIGS. **2**A and **2**B are schematic top views of an example façade system **200** that incorporates the principles of the present disclosure. The façade system **200** (hereafter “the system **200**”) may be similar in some respects to the system **100** of FIG. **1** and, therefore, may be best understood with reference thereto, where like numerals will represent like components not described again in detail. Similar to the system **100**, the system **200** may form part of a storefront system, but is equally applicable to other types of façade systems, such as curtain wall systems.

(25) As illustrated, the system **200** includes the vertical mullion **102** with the exterior and interior portions **104***a,b*, and the first and second panels **106***a,b* are secured to the mullion **102** using the exterior and interior gaskets **108***a,b*. Moreover, the mullion **102** includes the thermal break **112** arranged in the glazing pocket **110** and effectively dividing the glazing pocket **110** into the shallow and deep pockets **116***a,b*, as generally described above. It should be noted that while the mullion **102** is primarily described herein as a vertically-oriented member, embodiments are contemplated herein where the mullion **102** is installed as a horizontally-oriented member. In such embodiments, the principles of the present disclosure are equally applicable.

(26) Unlike the system **100** of FIG. **1**, however, the system **200** includes a collapsible element **202** arranged within the deep pocket **116***b*. In the illustrated embodiment, the collapsible element **202** extends between the mullion **102** and the adjacent lateral side **118** of the second panel **106***b*. More specifically, the collapsible element **202** extends between the lateral side **118** of the second panel **106***b* and the thermal break **112**, which forms part of the mullion **102**, as discussed above. In other embodiments, however, the collapsible element **202** could alternatively extend between other structural features of the deep pocket **116***b*, without departing from the scope of the disclosure.

(27) The collapsible element **202** may be made of a variety of materials including, but not limited to ethylene propylene diene terpolymer (EPDM), EPDM foam, foam rubber, thermoplastic vulcanisate (TPV), similar polymers, or any combination thereof.
(28) The collapsible element **202** is designed to be movable or collapsible between a collapsed state, as shown in FIG. **2**A, and an expanded state, as shown in FIG. **2**B. In some embodiments, the collapsible element **202** may be naturally biased to the expanded state, but could alternatively be naturally biased to the collapsed state. In some embodiments, the collapsible element **202** may be attached to and otherwise pre-assembled on the mullion **102** (e.g., attached to the thermal break **112**). In other embodiments, however, the collapsible element **202** may be attached to and otherwise pre-assembled on (attached to) the lateral side **118** of the second panel **106***b*.
(29) The collapsible element **202** is movable (transitionable) between the collapsed and expanded states during the assembly (installation) process of the second panel **106***b*. More particularly, in embodiments where the collapsible element **202** is naturally biased to the expanded state, advancing the second panel **106***b* into the deep pocket **116***b*, as generally described above, may cause the collapsible element **202** to collapse as the lateral side **118** of the second panel **106***b* approaches the thermal break **112**. Upon subsequently drawing or pulling the second panel **106***b* away from the thermal break **112** a small distance, as also generally described above, the collapsible element **202** may be allowed to expand back to (or at least partially to) the expanded state.
(30) In contrast, there may be embodiments where the collapsible element **202** is naturally biased to the collapsed state and pre-assembled (installed) on the thermal break **112** within the deep pocket **116***b*. In such embodiments, the second panel **106***b* may be advanced into the deep pocket **116***b* until engaging the lateral side **118** of the second panel **106***b* against the collapsible element **202** in the collapsed state. One or both of the lateral side **118** and the collapsible element **202** may have an adhesive or other coupling mechanism (e.g., Velcro) that attaches the collapsible element **202** to the lateral side **118** once the lateral side **118** contacts the collapsible element. Upon subsequently drawing (pulling) the second panel **106***b* away from the thermal break **112** a small distance within the deep pocket **116***b*, as generally described above, the collapsible element **202** may be pulled or urged to expand (at least partially) to the expanded state.
(31) As shown in FIG. **2**B, upon transitioning to the expanded state, the collapsible element **202** may divide the deep pocket **116***b* into two or more thermal chambers. In the illustrated embodiment, the expanded collapsible element **202** divides the deep pocket **116***b* into three thermal chambers, identified by the numbers “1”, “2”, and “3”. The multiple thermal chambers **1**, **2**, **3** divide the volume of air within the deep pocket **116***b* into fractions equal to the number of thermal chambers, which operates to reduce heat transfer by convection through the glazing pocket **110**.
(32) In the illustrated embodiment, the collapsible element **202** exhibits a design similar in some respects to an accordion or bellows. More particularly, the collapsible element **202** includes two side walls **204** designed and otherwise configured to fold (bend) inward upon moving to the collapsed state. Those skilled in the art will readily appreciate, however, that the collapsible element **202** may exhibit several different designs and configurations that are equally capable of transiting between the collapsed and expanded states, and equally capable of dividing the deep pocket **116***b* into a plurality of thermal chambers, without departing from the scope of the disclosure.
(33) It should be noted that the glazing pocket **110** where the collapsible element **202** is located is substantially sealed with the exterior and interior gaskets **108***a*,*b*. Consequently, the collapsible element **202** is not intended to operate as a type of gasket or otherwise perform a sealing function for the system **200**. Rather, the main function of the collapsible element **202**, as indicated above, is to reduce heat transfer by convection through the glazing pocket **110**. This same principle is applicable to the other collapsible element embodiments described herein.
(34) FIGS. **3**A and **3**B are schematic top views of another example façade system **300**, in

accordance with one or more additional embodiments of the present disclosure. The façade system **300** (hereafter "the system **300**") may be similar in some respects to the system **200** of FIGS. **2**A-**2**B and, therefore, may be best understood with reference thereto, where like numerals will represent like components not described again in detail. Similar to the system **200**, the system **300** may form part of a storefront system, but the principles of the present disclosure are equally applicable to other types of façade systems, such as curtain wall systems.

(35) As illustrated, the system **300** includes the mullion **102** with the exterior and interior portions **104***a,b*, and the first and second panels **106***a,b* secured to the mullion **102** using the exterior and interior gaskets **108***a,b*. Moreover, the mullion **102** includes the thermal break **112** arranged in the glazing pocket **110** and effectively dividing the glazing pocket **110** into the shallow and deep pockets **116***a,b*, as generally described above.

(36) The system **300** also includes a collapsible element **302** arranged within the deep pocket **116***b* and extending between the mullion **102** (e.g., the thermal break **112**) and the lateral side **118** of the second panel **106***b*. The collapsible element **302** may be similar in some respects to the collapsible element **202** of FIGS. **2**A-**2**B, and therefore may be best understood with reference thereto.

(37) The collapsible element **302** is movable or collapsible between a collapsed state, as shown in FIG. **3**A, and an expanded state, as shown in FIG. **3**B. In some embodiments, the collapsible element **302** may be naturally biased to the expanded state, but could alternatively be naturally biased to the collapsed state. In some embodiments, the collapsible element **302** may be attached to and otherwise pre-assembled on the mullion **102** (e.g., the thermal break **112**). In other embodiments, however, the collapsible element **302** may be attached to and otherwise pre-assembled on the lateral side **118** of the second panel **106***b*.

(38) The collapsible element **302** may be made of the same or similar materials as the collapsible element **202**, and may operate similarly during the assembly (installation) process.

(39) Upon transitioning to the expanded state, as shown in FIG. **3**B, the collapsible element **302** is designed to divide the deep pocket **116***b* into three thermal chambers, identified by the numbers "1", "2", and "3", which effectively divide the volume of air within the deep pocket **116***b* into smaller volumes and thereby reduces heat transfer by convection through the glazing pocket **110**. Similar to the collapsible element **202** of FIGS. **2**A-**2**B, the collapsible element **302** exhibits a design similar in some respects to an accordion or a bellows. In the illustrated embodiment, however, the collapsible element **302** includes two side walls **304** designed to fold (bend) outward upon moving to the collapsed state.

(40) FIGS. **4**A and **4**B are schematic top views of another example façade system **400** in accordance with one or more additional embodiments of the present disclosure. The façade system **400** (hereafter "the system **400**") may be similar in some respects to the façade systems **200** and **300** of FIGS. **2**A-**2**B and **3**A-**3**B and, therefore, may be best understood with reference thereto. Similar to the systems **200** and **300**, the system **400** includes a collapsible element **402** arranged within the deep pocket **116***b* and extending between the mullion **102** (e.g., the thermal break **112**) and the lateral side **118** of the second panel **106***b*.

(41) The collapsible element **402** may be similar in some respects to the collapsible elements **202** and **302** of FIGS. **2**A-**2**B and **3**A-**3**B, and therefore may be best understood with reference thereto. The collapsible element **402** is movable (collapsible) between a collapsed state, as shown in FIG. **4**A, and an expanded state, as shown in FIG. **4**B. In some embodiments, the collapsible element **402** may be naturally biased to the expanded state, but could alternatively be naturally biased to the collapsed state. In some embodiments, the collapsible element **402** may be attached to and otherwise pre-assembled on the mullion **102** (e.g., the thermal break **112**). In other embodiments, however, the collapsible element **402** may be attached to and otherwise pre-assembled on the lateral side **118** of the second panel **106***b*.

(42) The collapsible element **402** may be made of the same or similar materials as the collapsible element **202**, and may operate similarly during the assembly (installation) process.

(43) Upon transitioning to the expanded state, as shown in FIG. **4**B, the collapsible element **402** is designed to divide the deep pocket **116***b* into four thermal chambers, identified by the numbers “1”, “2”, “3”, and “4”, which effectively divide the volume of air within the deep pocket **116***b* into corresponding fractions that reduce heat transfer by convection through the glazing pocket **110**.
(44) Similar to the collapsible elements **202** and **302** of FIGS. **2**A-**2**B and **3**A-**3**B, the collapsible element **402** exhibits a design similar in some respects to an accordion or bellows. In contrast to the collapsible elements **202** and **302**, however, the collapsible element **402** includes three walls that divide the deep pocket **116***b* into the four thermal chambers **1**, **2**, **3**, **4**. More specifically, the collapsible element **402** provides opposing side walls **404***a* and **404***b*, and an inner wall **406** interposing the side walls **404***a,b*. The side walls **404***a,b* are configured to exhibit an exterior fold (i.e., fold outward), while the inner wall **406** exhibits a fold directed either inward or outward and toward one side or the other upon moving to the collapsed state.
(45) FIGS. **5**A and **5**B are schematic top views of another example façade system **500**, in accordance with one or more additional embodiments of the present disclosure. The façade system **500** (hereafter “the system **500**”) may be similar in some respects to the façade systems **200**, **300**, and **400** described above and, therefore, may be best understood with reference thereto. Similar to the systems **200**, **300**, and **400**, the system **500** includes a collapsible element **502** arranged within the deep pocket **116***b* and extending between the mullion **102** (e.g., the thermal break **112**) and the lateral side **118** of the second panel **106***b*.
(46) The collapsible element **502** may be similar in some respects to the collapsible elements **202**, **302**, and **402** described above, and therefore may be best understood with reference thereto. The collapsible element **502** is movable (collapsible) between a collapsed state, as shown in FIG. **5**A, and an expanded state, as shown in FIG. **5**B. In some embodiments, the collapsible element **502** may be naturally biased to the expanded state, but could alternatively be naturally biased to the collapsed state. In some embodiments, as illustrated, the collapsible element **502** may be attached to and otherwise pre-assembled on the mullion **102** (e.g., the thermal break **112**). In other embodiments, however, the collapsible element **502** may be attached to and otherwise pre-assembled on the lateral side **118** of the second panel **106***b*.
(47) The collapsible element **502** may be made of the same or similar materials as the collapsible element **202**, and may operate similarly during the assembly (installation) process.
(48) Upon transitioning to the expanded state, as shown in FIG. **5**B, the collapsible element **502** is designed to divide the deep pocket **116***b* into two thermal chambers, identified by the numbers “1” and “2”, which effectively divide the volume of air within the deep pocket **116***b* and thereby reduce heat transfer by convection through the glazing pocket **110**. The collapsible element **502** includes two side walls **504***a* and **504***b* and a cross-member **506** extending between the two side walls **504***a,b*. When the collapsible element **502** is in the collapsed state, the side walls **504***a,b* may be folded over one another. Upon transitioning to the expanded state, however, at least one of the side walls **504***a,b* may extend to the lateral side **118** of the second panel **106***b*.
(49) FIGS. **6**A and **6**B are schematic top views of another example façade system **600**, in accordance with one or more additional embodiments of the present disclosure. The façade system **600** (hereafter “the system **600**”) may be similar in some respects to the façade systems **200**, **300**, **400**, and **500** described above and, therefore, may be best understood with reference thereto. Similar to the systems **200**, **300**, **400**, and **500**, for example, the system **600** includes a collapsible element **602** arranged within the deep pocket **116***b* and extending between the mullion **102** and the second panel **106***b*.
(50) The collapsible element **602** may be similar in some respects to the collapsible elements **202**, **302**, **402**, and **502** described above, and therefore may be best understood with reference thereto. The collapsible element **602** is movable (collapsible) between a collapsed state, as shown in FIG. **6**A, and an expanded state, as shown in FIG. **6**B. In some embodiments, the collapsible element **602** may be naturally biased to the expanded state, but could alternatively be naturally biased to the

collapsed state.
(51) As illustrated, the collapsible element **602** may comprise multiple portions, shown as a first or "exterior" portion **604***a* and a second or "interior" portion **604***b* separate from the exterior portion **604***a*. The portions **604***a*,*b* may be attached to and otherwise pre-assembled on the mullion **102** prior to installation of the second panel **106***b*. More specifically, each portion **604***a*,*b* provides a side wall **606** interconnected with a foldable inner wall **608**. The side walls **606** may be secured to adjacent inner portions of the mullion **102** and extend substantially parallel with the exterior and interior exposed surfaces **610***a* and **610***b* of the second panel **106***b*.
(52) In contrast, the foldable inner walls **608** may extend from the corresponding side wall **606** at a living hinge and be able to flex or pivot between the collapsed and expanded states. When in the collapsed state, the inner walls **608** may interpose the thermal break **112** and the lateral side of the second panel **106***b*. Upon transitioning to the expanded state, however, the inner walls **608** may be configured to flex away from the thermal break **112**. In some embodiments, the end of each inner wall **608** may engage the lateral side **118** of the second panel **106***b* when transitioned to the expanded state.
(53) When transitioned to the expanded state, the collapsible element **602** may be configured to divide the deep pocket **116***b* into three thermal chambers, identified by numbers "1", "2", and "3", which divide the volume of air within the deep pocket **116***b* and thereby reduce heat transfer by convection through the glazing pocket **110**.
(54) FIGS. **7**A and **7**B are schematic top views of another example façade system **700**, in accordance with one or more additional embodiments of the present disclosure. The façade system **700** (hereafter "the system **700**") may be similar in some respects to the façade systems **200**, **300**, **400**, **500**, and **600** described above and, therefore, may be best understood with reference thereto. Similar to the systems **200**, **300**, **400**, **500**, and **600**, the system **700** includes a collapsible element **702** arranged within the deep pocket **116***b* and extending or extendible between the mullion **102** (e.g., the thermal break **112**) and the lateral side **118** of the second panel **106***b*.
(55) The collapsible element **702** may be similar in some respects to the collapsible elements **202**, **302**, **402**, **502**, and **602** described above, and therefore may be best understood with reference thereto. The collapsible element **702** is movable (collapsible) between a collapsed state, as shown in FIG. **7**A, and an expanded state, as shown in FIG. **7**B. In some embodiments, the collapsible element **702** may be naturally biased to the expanded state, but could alternatively be naturally biased to the collapsed state.
(56) As best seen in FIG. **7**B, the collapsible element **702** may include opposing side walls **704** secured to and otherwise arranged adjacent opposing inner portions of the mullion **102**. The side walls **704** may extend substantially parallel with the exterior and interior exposed surfaces **610***a* and **610***b* of the second panel **106***b*. The collapsible element **702** may further include a cross-member **706** extending between and interconnecting the opposing side walls **704**. As illustrated, the cross-member **706** may be secured to and otherwise arranged adjacent the thermal break **112**.
(57) The collapsible element **702** may further include one or more foldable inner walls **708** (two shown) that are able to transition between the collapsed and expanded states. More specifically, each inner wall **708** extends from a transition point where the sidewalls **704** meet the cross-member **706**. When in the collapsed state, the inner walls **708** may interpose the cross-member **706** and the lateral side of the second panel **106***b*. Upon transitioning to the expanded state, however, the inner walls **708** may be configured to flex away from the cross-member **706**. In some embodiments, the end of each inner wall **708** may engage the lateral side **118** of the second panel **106***b* when transitioned to the expanded state.
(58) Upon transitioning to the expanded state, as shown in FIG. **7**B, the collapsible element **702** is designed to divide the deep pocket **116***b* into three thermal chambers, identified by the numbers "1", "2", and "3", which divide the volume of air within the deep pocket **116***b* and thereby reduce heat transfer by convection through the glazing pocket **110**.

(59) As mentioned herein, the presently disclosed collapsible elements may be attached to and otherwise pre-assembled on the mullion **102** or alternatively on the lateral side **118** of the second panel **106***b*. FIGS. **8-11** depict example attachment means for securing collapsible elements within the corresponding systems.
(60) FIG. **8** shows the collapsible element **202** of FIGS. **2**A-**2**B secured within the system **200** using an adhesive **802**. In the illustrated embodiment, the adhesive **802** interposes the collapsible element **202** and a portion of the mullion **102**, such as the thermal break **112**. In such embodiments, the collapsible element **202** may be pre-installed within the deep pocket **116***b*. In other embodiments, however, the adhesive **802** may interpose the collapsible element **202** and the lateral side **118** of the second panel **106***b*. In such embodiments, the collapsible element **202** may be pre-installed on the second panel **106***b*. In yet other embodiments, the adhesive may be applied at both interfaces between the collapsible gasket **202** and the thermal break **112**, and between the collapsible gasket **202** and the lateral side **118** of the second panel **106***b*. Upon drawing the second panel **106***b* partially out of the deep pocket **116***b*, as described herein, the collapsible element **202** may be urged to expand.
(61) FIG. **9** shows the collapsible element **202** of FIGS. **2**A-**2**B secured within the system **200** using a coupling device **902**. In the illustrated embodiment, the coupling device **902** comprises a snap-fit or tongue-and-groove attachment coupled to the collapsible element **202** and capable of being secured to the bridge **114** forming part of the thermal break **112**. More specifically, the coupling device **902** may provide or otherwise define a head **904** receivable within an aperture or channel **906** defined in the bridge **114**.
(62) In some embodiments, the collapsible element **202** may be pre-installed within the deep pocket **116***b* and attached to the bridge **114**. In such embodiments, the head **904** may be received within the channel **906**, and introducing the second panel **106***b* into the deep pocket **116***b* will compress the collapsible element **202**, but drawing the second panel **106***b* partially out of the deep pocket **116***b* will allow the collapsible element **202** to expand. In other embodiments, however, the collapsible element **202** may be pre-installed on and otherwise secured to the lateral side **118** of the second panel **106***b*. In such embodiments, introducing the second panel **106***b* into the deep pocket **116***b* will allow the head **904** to locate and be received within the channel **906** as the collapsible element **202** is compressed. Once the coupling device **902** is secured to the bridge **114**, drawing the second panel **106***b* partially out of the deep pocket **116***b* will allow the collapsible element **202** to expand.
(63) FIG. **10** shows the collapsible element **202** of FIGS. **2**A-**2**B secured within the system **200** using an alternative type of coupling device **1002**. In the illustrated embodiment, the coupling device **1002** comprises a snap-fit or interference-fit attachment configured to be secured to thermal break **112**, such as the bridge **114**.
(64) In some embodiments, the collapsible element **202** may be pre-installed within the deep pocket **116***b* and attached to the thermal break **112** (e.g., the bridge **114**) using the coupling device **1002**. In other embodiments, however, the collapsible element **202** may be pre-installed on and otherwise secured to the lateral side **118** of the second panel **106***b*. In such embodiments, introducing the second panel **106***b* into the deep pocket **116***b* will allow the coupling device **1002** to engage and become secured to the thermal break **112** (e.g., the bridge **114**) as the collapsible element **202** is compressed. Once the coupling device **1002** is secured to the thermal break **112**, drawing the second panel **106***b* partially out of the deep pocket **116***b* will allow the collapsible element **202** to expand.
(65) FIG. **11** shows the collapsible element **702** of FIGS. **7**A-**7**B secured within the system **700**. In the illustrated embodiment, the collapsible element **702** may include a coupling device **1102** configured to secure the collapsible element **702** to the mullion **102** via an interference fit or a snap-fit engagement. More specifically, the opposing side walls **704** of the collapsible element **702** may comprise the coupling device **1102**, and may be sized and otherwise configured to form a

snap-fit or interference-fit engagement with corresponding grooves **1104** defined by the mullion **102** within the glazing pocket **110** (e.g., the deep pocket **116***b*). Similar to the side walls **704**, the grooves **1104** may extend substantially parallel with the exterior and interior exposed surfaces **610***a* and **610***b* of the second panel **106***b*. The collapsible element **702** may be pre-installed within the deep pocket **116***b*.

(66) FIG. **12** is a top view of an example façade system **1200** that may incorporate the principles of the present disclosure. In the illustrated embodiment, the façade system **1200** (hereafter the "system **1200**") comprises a curtain wall assembly configured to help laterally support and/or secure the first and second panels **106***a*,*b*. As illustrated, the system **1200** may include a vertical mullion **1202**, which may comprise a rigid extrusion made of aluminum, an aluminum alloy, or other material, including, but not limited to, other metals and alloys. The vertical mullion **1202** may be coupled to a building structure, such as a beam that forms part of the building structure.

(67) The system **1200** may further include a pressure plate **1204** and a cover **1206** removably coupled to the pressure plate **1204**. The pressure plate **1204** may be operatively coupled to the vertical mullion **1202** with a fastener **1208**, which may be a mechanical fastener, that extends through a glazing pocket **1210** defined laterally between the vertical mullion **1202** and the pressure plate **1204**, and defined horizontally between the first and second glazing panels **106***a*,*b*. In the illustrated embodiment, the fastener **1208** comprises a screw that may be received within or otherwise threaded into a tongue **1212** extending from or forming part of the vertical mullion **1202**. The system **1200** may further include a thermal separator **1214** positioned within the glazing pocket **1210** and interposing the pressure plate **1204** and the vertical mullion **1202** (e.g., the tongue **1212**).

(68) The system **1200** may further include one or more collapsible elements arranged within the glazing pocket **1210**. In the illustrated embodiment, a first collapsible element **1216***a* is arranged in the glazing pocket **1210** and interposes the tongue **1212** and a lateral end **1218** of the first panel **106***a*. A second collapsible element **1216***b* is also arranged in the glazing pocket **1210**, but interposes the tongue **1212** and the lateral end **118** of the second panel **106***b*. The collapsible elements **1216***a*,*b* are movable (collapsible) between collapsed and expanded states during installation of the system **1200**. In some embodiments, the collapsible elements **1216***a*,*b* may be naturally biased to the expanded state, but could alternatively be naturally biased to the collapsed state. In some embodiments, the collapsible elements **1216***a*,*b* may be attached to and otherwise pre-assembled on the mullion **1202** (e.g., the tongue **1212**), but could alternatively be attached to and otherwise pre-assembled on the lateral sides **1218**, **118** of one or both of the panels **106***a*,*b*.

(69) In the illustrated embodiment, the collapsible elements **1216***a*,*b* are the same as or similar to the collapsible element **202** of FIGS. **2**A-**2**B. Accordingly, upon transitioning to the expanded state, as shown in FIG. **12**, the collapsible elements **1216***a*,*b* may be designed to divide corresponding portions of the glazing pocket **1210** into three thermal chambers, identified by the numbers "1", "2", and "3". The multiple thermal chambers **1**,**2**,**3** divide the volume of air within the glazing pocket **1210**, which operates to reduce heat transfer by convection through the glazing pocket **1210**. In other embodiments, however, the collapsible elements **1216***a*,*b* may be replaced with any of the collapsible elements described herein, without departing from the scope of the disclosure.

(70) FIGS. **13**A and **13**B are side-by-side depictions of thermal simulations of the system **200** of FIGS. **2**A-**2**B. More specifically, FIG. **13**A depicts the system **200** without a collapsible element, and FIG. **13**B depicts the system **200** including the collapsible element **202**, as generally described above with reference to FIGS. **2**A-**2**B. The thermal simulations were performed using commercially-available heat transfer software.

(71) Table 1 below provides testing data comparing a conventional vertical mullion system without a collapsible element, to a vertical mullion system that includes a collapsible element, as generally described herein. It can be seen that the U-factor of the system provided with the thermal element is lower, therefore providing a better thermal performance and energy savings.

(72) TABLE-US-00001 TABLE 1 Improvement System U-factor U-factor [%] Conventional

vertical 0.8795 4.9943 W/m.sup.2-K — mullion w/o collapsible Btu/h-ft.sup.2-F element Vertical mullion w/ 0.8122 4.6121 W/m.sup.2-K 8.3% collapsible element Btu/h-ft.sup.2-F

(73) Therefore, the disclosed systems and methods are well adapted to attain the ends and advantages mentioned as well as those that are inherent therein. The particular embodiments disclosed above are illustrative only, as the teachings of the present disclosure may be modified and practiced in different but equivalent manners apparent to those skilled in the art having the benefit of the teachings herein. Furthermore, no limitations are intended to the details of construction or design herein shown, other than as described in the claims below. It is therefore evident that the particular illustrative embodiments disclosed above may be altered, combined, or modified and all such variations are considered within the scope of the present disclosure. The systems and methods illustratively disclosed herein may suitably be practiced in the absence of any element that is not specifically disclosed herein and/or any optional element disclosed herein. While compositions and methods are described in terms of "comprising," "containing," or "including" various components or steps, the compositions and methods can also "consist essentially of" or "consist of" the various components and steps. All numbers and ranges disclosed above may vary by some amount. Whenever a numerical range with a lower limit and an upper limit is disclosed, any number and any included range falling within the range is specifically disclosed. In particular, every range of values (of the form, "from about a to about b," or, equivalently, "from approximately a to b," or, equivalently, "from approximately a-b") disclosed herein is to be understood to set forth every number and range encompassed within the broader range of values. Also, the terms in the claims have their plain, ordinary meaning unless otherwise explicitly and clearly defined by the patentee. Moreover, the indefinite articles "a" or "an," as used in the claims, are defined herein to mean one or more than one of the elements that it introduces. If there is any conflict in the usages of a word or term in this specification and one or more patent or other documents that may be incorporated herein by reference, the definitions that are consistent with this specification should be adopted.

(74) As used herein, the phrase "at least one of" preceding a series of items, with the terms "and" or "or" to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase "at least one of" allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases "at least one of A, B, and C" or "at least one of A, B, or C" each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

(75) Although various example embodiments have been disclosed, a worker of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. For that reason, the following claims should be studied to determine the scope and content of this disclosure.

Claims

1. A façade system, comprising: a mullion having exterior and interior portions and defining a glazing pocket between the exterior and interior portions; a thermal break arranged within the glazing pocket and extending between the exterior and interior portions, the thermal break dividing the glazing pocket into a shallow pocket and a deep pocket larger than the shallow pocket; and a collapsible element arranged within the deep pocket and extending between the thermal break and a lateral side of a panel introduced into the deep pocket, the collapsible element being movable between a collapsed state and an expanded state and including: a first portion engageable with the thermal break; a second portion opposite the first portion and engageable with the lateral side of the panel; and opposing first and second side walls extending between the first and second portions and being foldable as the collapsible element transitions between the expanded and collapsed states, and wherein an interior surface of the first and second side walls is engageable with the first and

second portions, respectively, when the collapsible element is moved to the collapsed state and an expanded state, and wherein the collapsible element divides the deep pocket into three thermal chambers when in the expanded state to reduce heat transfer by convection through the glazing pocket.

2. The façade system of claim 1, wherein the collapsible element is naturally biased to the expanded state.

3. The façade system of claim 1, wherein the collapsible element is naturally biased to the collapsed state.

4. The façade system of claim 1, wherein the first and second side walls fold inward upon moving to the collapsed state.

5. The façade system of claim 1, wherein the first and second side walls fold outward upon moving to the collapsed state.

6. The façade system of claim 1, wherein the collapsible element further includes an inner wall interposing the two side walls, and wherein the two side walls and the inner wall divide the deep pocket into four thermal chambers.

7. The façade system of claim 6, wherein the two side walls fold outward and the inner wall folds toward one of the two side walls upon moving to the collapsed state.

8. The façade system of claim 1, wherein the collapsible element is secured to the mullion or the panel with an attachment means selected from the group consisting of an adhesive, a coupling device, an interference fit, a snap-fit engagement, and any combination thereof.

9. The façade system of claim 1, wherein the panel comprises a first panel and the system further comprises: a second panel laterally offset from the first panel and received within the shallow pocket, wherein the glazing pocket is defined between the exterior and interior portions of the mullion and between lateral ends of the first and second panels; first and second exterior gaskets providing corresponding sealed interfaces between the first and second panels and the exterior portion of the mullion; and first and second interior gaskets providing corresponding sealed interfaces between the first and second panels and the interior portion of the mullion, wherein the first and second exterior and interior gaskets substantially seal the glazing pocket.

10. A method of assembling a façade system, comprising: coupling a first panel to a mullion, the mullion including: an exterior portion and an interior portion; a glazing pocket defined between the exterior and interior portions; and a thermal break arranged within the glazing pocket and extending between the exterior and interior portions and thereby dividing the glazing pocket into a shallow pocket and a deep pocket larger than the shallow pocket, wherein the first panel is received within the shallow pocket; advancing a second panel into the deep pocket and toward the thermal break, wherein a collapsible element is arranged in the deep pocket and includes: a first portion engageable with the thermal break; a second portion opposite the first portion and engageable with a lateral side of the second panel; and opposing first and second side walls extending between the first and second portions and being foldable as the second panel advances into the deep pocket; transitioning the collapsible element between a collapsed state and an expanded state as the second panel is received within the deep pocket, wherein an interior surface of the first and second side walls is engageable with the first and second portions, respectively, when the collapsible element is moved to the collapsed state; and dividing the deep pocket into three thermal chambers with the collapsible element in the expanded state.

11. The method of claim 10, wherein the collapsible element is naturally biased to the expanded state and advancing the second panel into the deep pocket comprises collapsing the collapsible element to the collapsed state as the second panel advances into the deep pocket.

12. The method of claim 11, further comprising advancing the second panel into the second pocket at an angle offset from perpendicular to the thermal break.

13. The method of claim 10, further comprising drawing the second panel partially out of the deep pocket and thereby allowing the collapsible element to transition from the collapsed state to the

expanded state.

14. The method of claim 10, wherein the collapsible element is secured to at least one of the thermal break and the lateral side of the panel with an attachment means selected from the group consisting of an adhesive, a coupling device, an interference fit, a snap-fit engagement, and any combination thereof.

15. A method of reducing heat transfer through the façade system of claim 1, the method comprising: dividing the deep pocket of the glazing pocket into the three thermal chambers with the collapsible element when the collapsible element is transitioned to the expanded state; and reducing heat transfer by convection through the glazing pocket with the collapsible element in the expanded state.

16. The façade system of claim 1, further comprising a coupling device securing the collapsible element to the thermal break, wherein the coupling device includes a head inserted into a bridge of the thermal break.

17. A façade system, comprising: a mullion having exterior and interior portions and defining a glazing pocket between the exterior and interior portions; a thermal break arranged within the glazing pocket and extending between the exterior and interior portions, the thermal break dividing the glazing pocket into a shallow pocket and a deep pocket larger than the shallow pocket; and a collapsible element arranged within the deep pocket and extending between the thermal break and a lateral side of a panel introduced into the deep pocket, the collapsible element being movable between a collapsed state and an expanded state and including: a first portion engageable with the thermal break; a second portion opposite the first portion engageable with the lateral side of the panel; and opposing first and second side walls extending between the first and second portions that fold outward as the collapsible element transitions from the expanded state to the collapsed state, wherein collapsible element divides the deep pocket into three or more thermal chambers when the collapsible element is in the expanded state to reduce heat transfer by convection through the glazing pocket.

18. The façade system of claim 17, wherein the first portion and second portion contact one another when the collapsible element is in the collapsed state.

19. The façade system of claim 17, wherein the collapsible element is naturally biased to the collapsed state.

20. The façade system of claim 17, the collapsible element further comprising an inner wall interposing the first and second side walls that folds toward one of the first and second side walls as the collapsible element moves from the expanded state to the collapsed state.